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Database schema reference

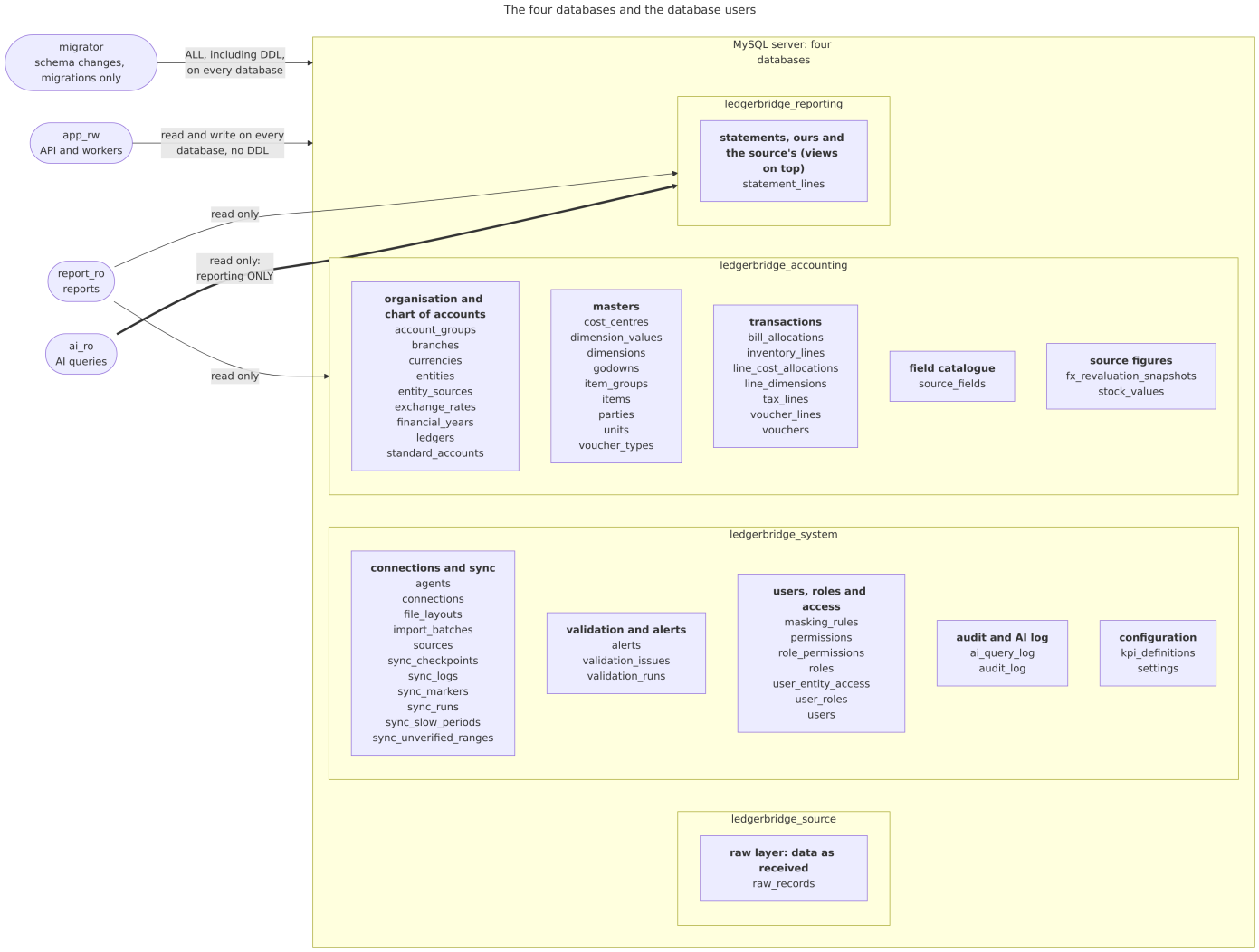
Every database, table and column of the client-hosted data plane (MySQL 8), generated from the code. For the design and its reasons see schema.md and the decision log in architecture.md; for the conversion and validation rules see the rulebook.

55 tables, 602 columns.

Conventions. Amounts are `DECIMAL(20,4)` in the base currency with the **canonical sign**: debit positive, credit negative (schema.md 1, 7.1). Every accounting table carries its source tracking columns (`connection_id`, `source_key`, `source_alter_id`, `origin`, `import_batch_id`, `raw_record_id`). Rows deleted in the source are kept with `is_deleted` (soft delete). `PK` primary key, `FK` foreign key, `UK` part of a unique key.

1. The four databases

Database	What it holds	Tables
`ledgerbridge_source`	Raw layer: every record exactly as the source sent it (ADR-005). Written only by connectors through the sync; the transform reads it.	1
`ledgerbridge_accounting`	Canonical model: companies, chart of accounts, masters and transactions in one shape for every source. Written by the transform; read by validation, reports and (later) the reporting layer.	28
`ledgerbridge_reporting`	Semantic layer for reports and AI (M6): views and aggregate tables. Empty until M6; the only database the AI user can read (AI-003).	1
`ledgerbridge_system`	Running the platform: connections, sync runs and progress, validation, alerts, users, roles, audit and AI logs, configuration.	25



2. Database users and their rights

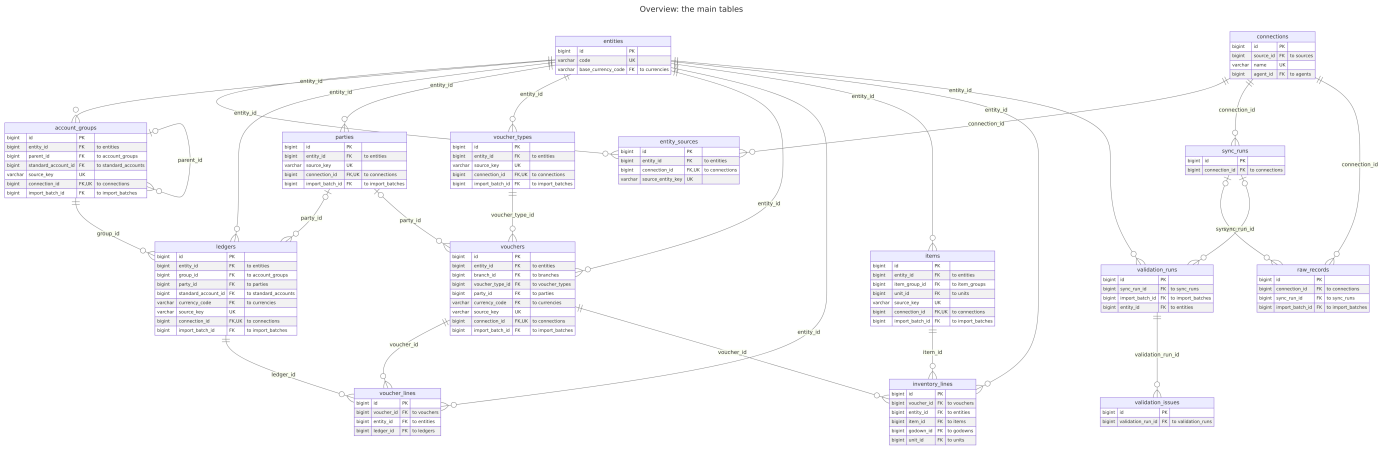
Created by `deploy/mysql/init/01-schemas.sh` (schema.md 2.1). Each user has only the rights it needs; the AI can read the reporting layer and nothing else.

User	Used by	Rights	Databases
`migrator`	schema changes, migrations only	ALL, including DDL, on every database	`ledgerbridge_source`, `ledgerbridge_system`, `ledgerbridge_accounting`, `ledgerbridge_reporting`
`app_rw`	API and workers	read and write on every database, no DDL	`ledgerbridge_source`, `ledgerbridge_system`, `ledgerbridge_accounting`, `ledgerbridge_reporting`
`report_ro`	reports	read only	`ledgerbridge_accounting`, `ledgerbridge_reporting`
`ai_ro`	AI queries	read only: reporting ONLY	`ledgerbridge_reporting`

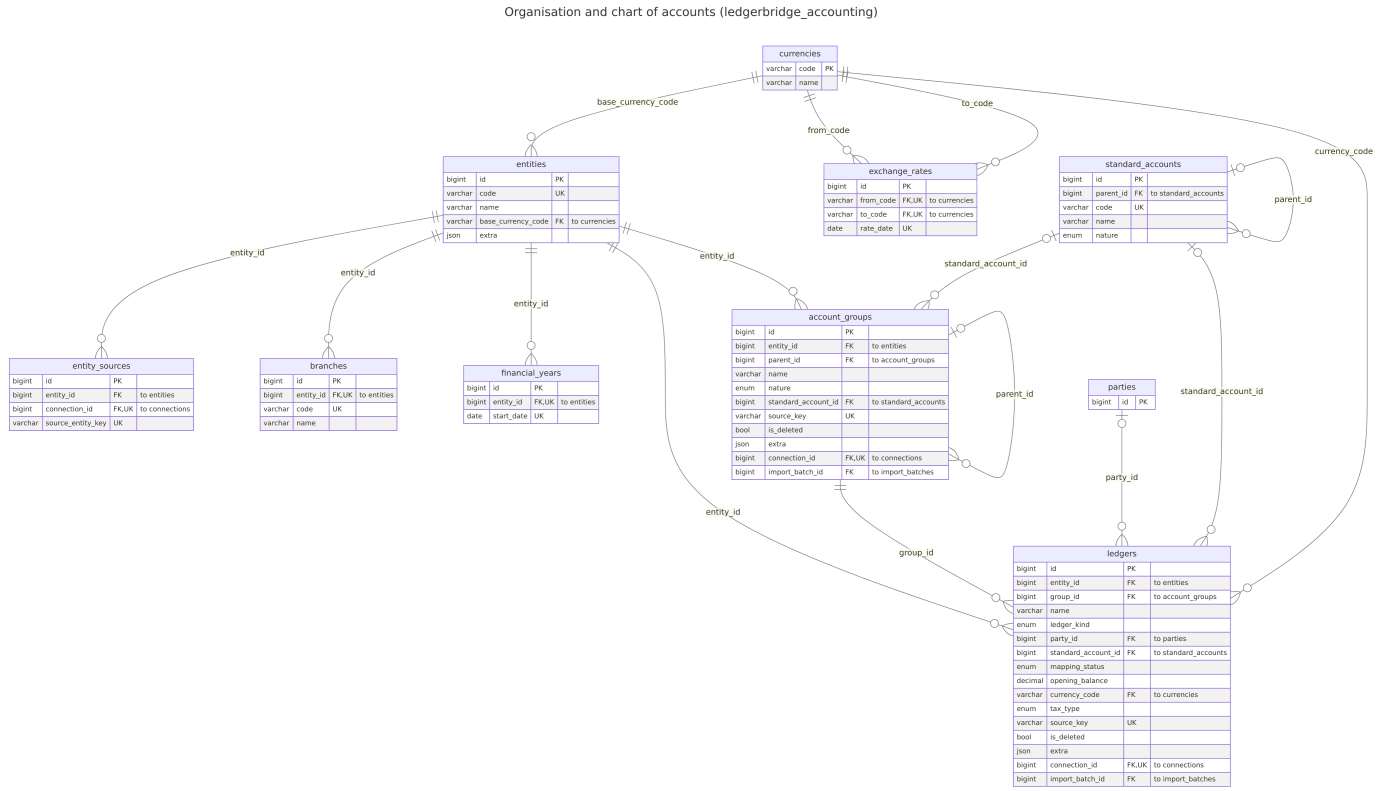
3. Diagrams

How to read them: docs/diagrams/README.md.

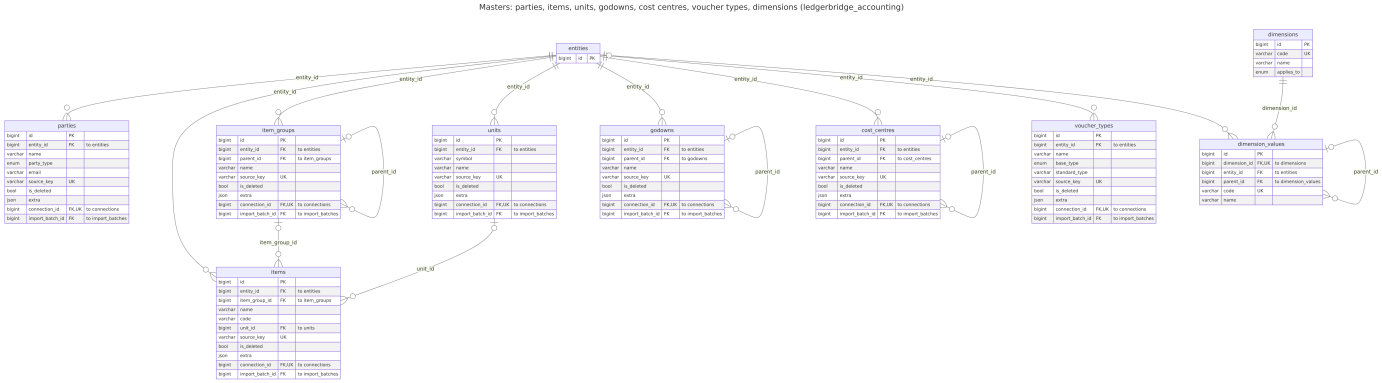
Overview (keys only)



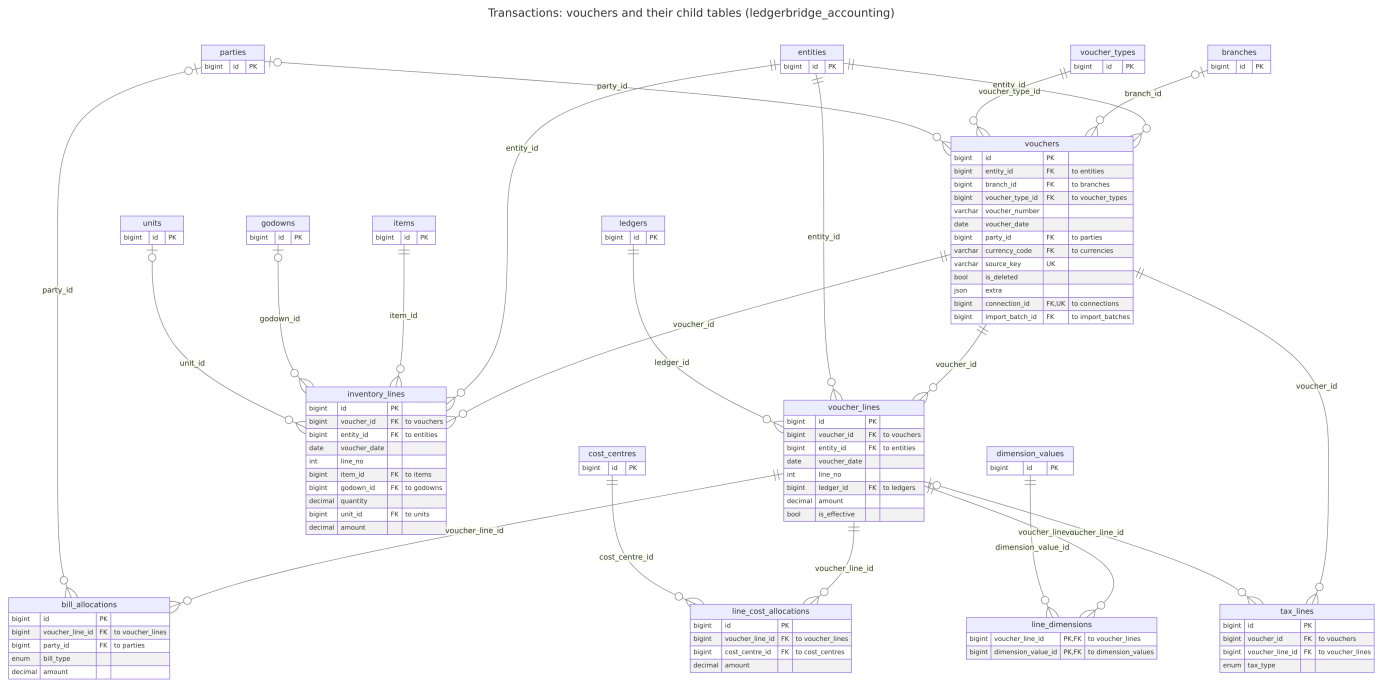
Organisation and chart of accounts (ledgerbridge_accounting)



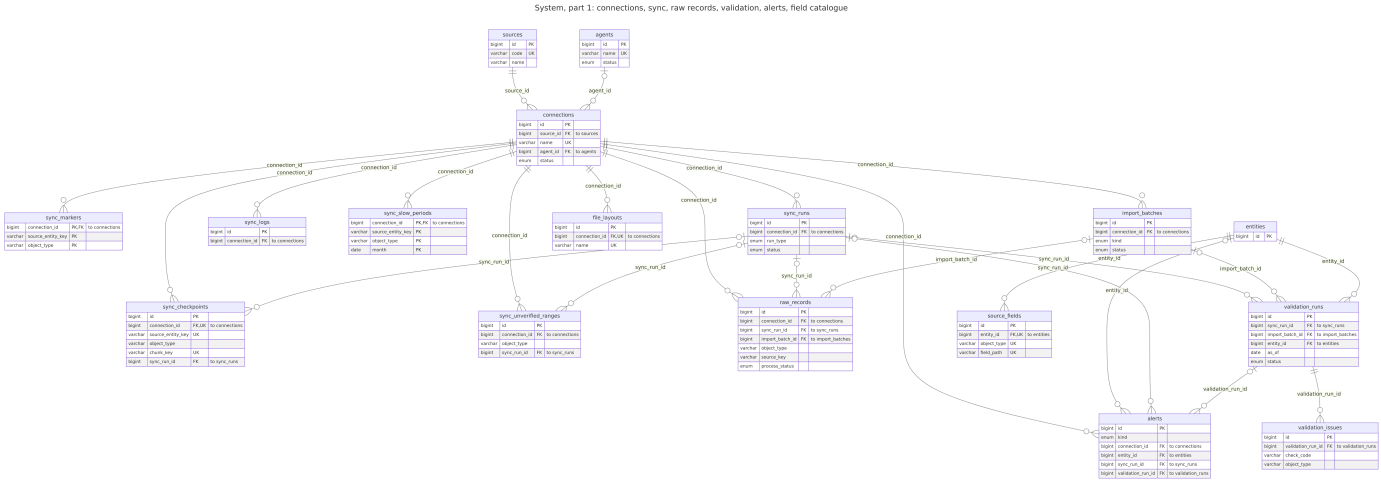
Masters: parties, items, units, godowns, cost centres, voucher types, dimensions (ledgerbridge_accounting)



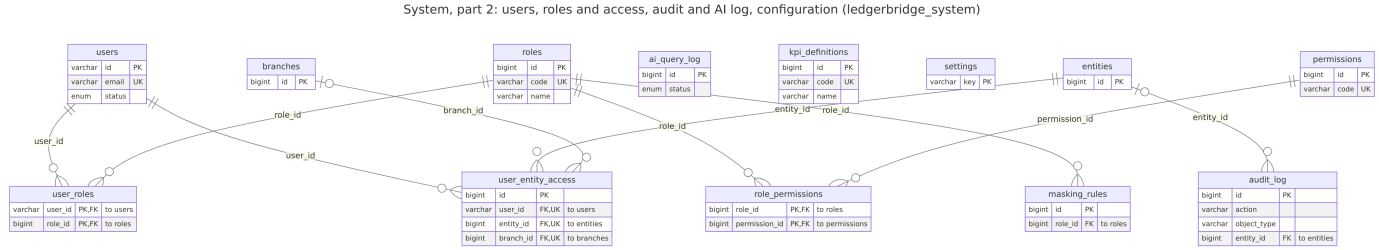
Transactions: vouchers and their child tables (ledgerbridge_accounting)



System, part 1: connections, sync, raw records, validation, alerts, field catalogue

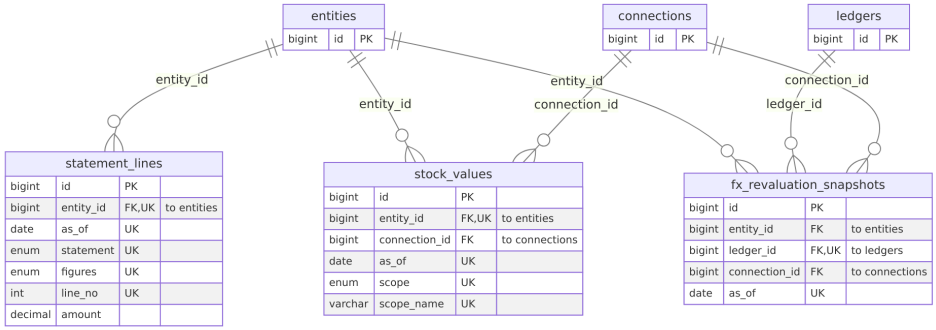


System, part 2: users, roles and access, audit and AI log, configuration (ledgerbridge_system)

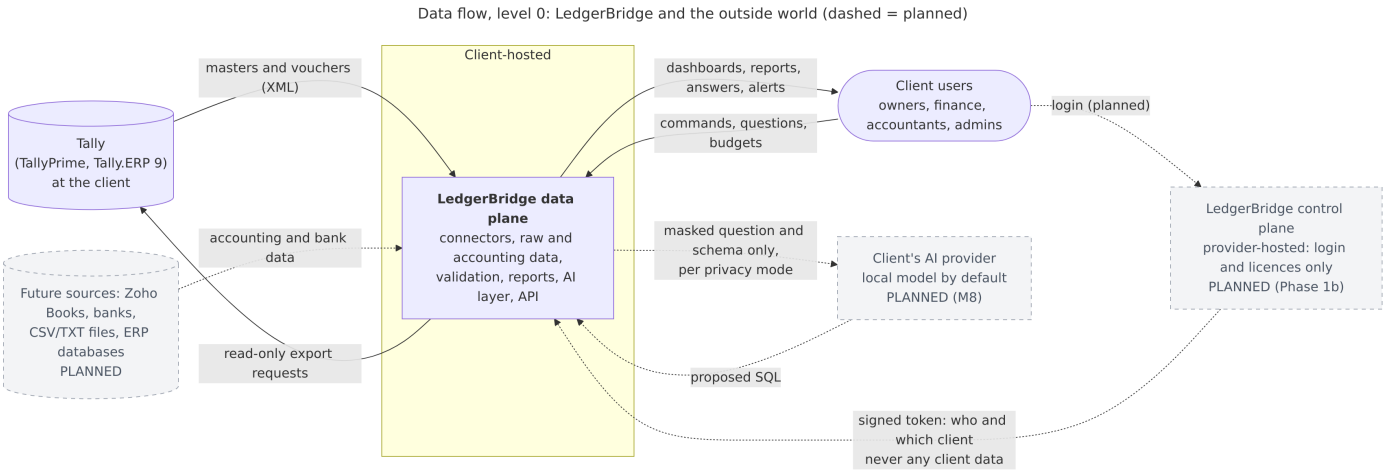


Source figures and statements: stock values, foreign-currency revaluations (ledgerbridge_accounting) and statement lines (ledgerbridge_reporting)

Source figures and statements: stock values, foreign-currency revaluations (ledgerbridge_accounting) and statement lines (ledgerbridge_reporting)

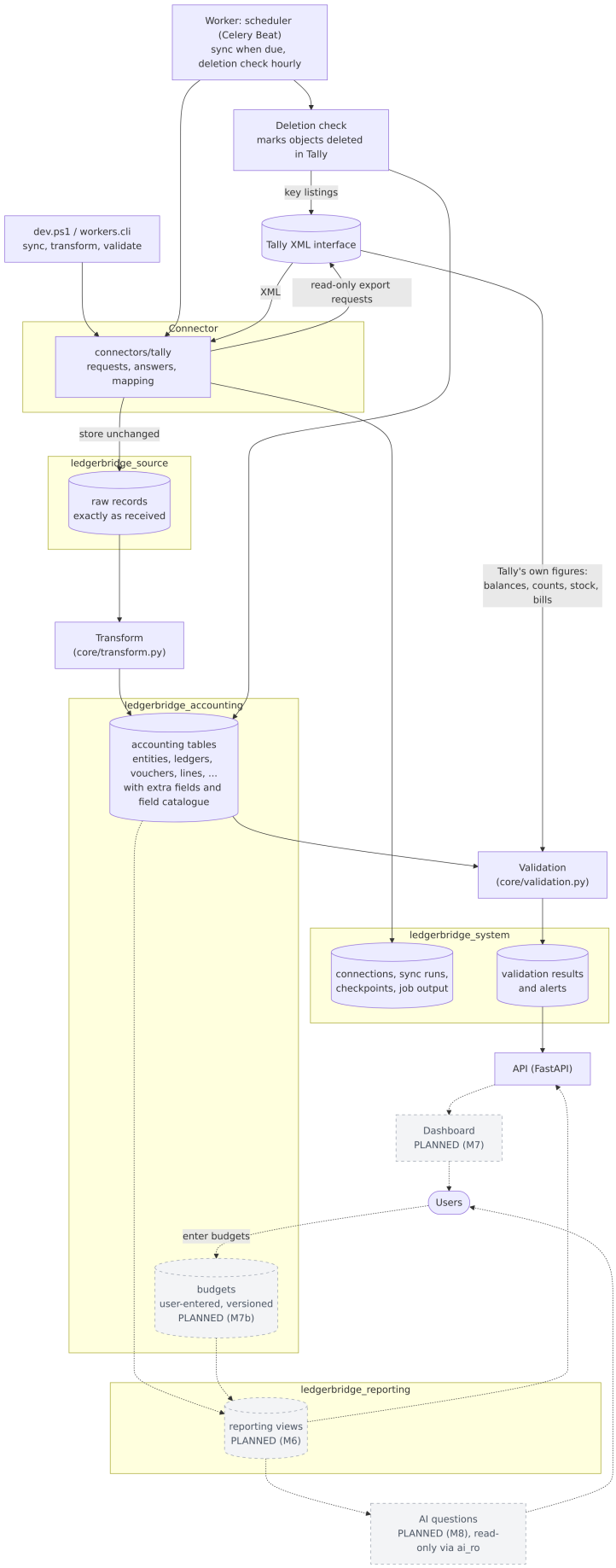


Data flow, level 0 (the platform and what it talks to)



Data flow, level 1 (inside the data plane)

Data flow, level 1: inside the data plane (dashed = planned)



4. Tables

ledgerbridge_source: raw layer: data as received

ledgerbridge_source.raw_records

Purpose: Every object fetched from a source, exactly as received (XML for Tally), one row per version.

Filled by: Sync (connector -> RawStore): a new row only when the payload changed (hash); process_status tracks the transform.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.
sync_run_id	BIGINT UNSIGNED	yes	FK	The sync run it belongs to.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
source_entity_key	VARCHAR(191)	no		The source company's key (Tally: company GUID).
object_type	VARCHAR(64)	no		group, ledger, voucher_type, voucher, stock_item, …
source_key	VARCHAR(191)	no		The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
payload_format	ENUM('xml','json','csv','text')	no		Payload format.
payload	LONGTEXT	no		The record exactly as received (re-serialised XML).
payload_hash	CHAR(64)	no		SHA-256 of the payload: an unchanged record isn't stored again.
received_at	DATETIME(6)	no		When: received.
process_status	ENUM('pending','processed','failed','skipped')	no		pending / processed / failed / skipped (superseded).
process_error	TEXT	yes		Why the transform failed: error type and a short message, never SQL.
process_warning	TEXT	yes		Process warning.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index ix_raw_records_conn_type_key (connection_id, object_type, source_key); index ix_raw_records_status_received (process_status, received_at).

Refers to: connection_id → connections.id; sync_run_id → sync_runs.id; import_batch_id → import_batches.id.

ledgerbridge_accounting: organisation and chart of accounts

ledgerbridge_accounting.entities

Purpose: A company in the platform (one per source company).

Filled by: Transform, from the source's company record, on the first sync.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(32)	no	UK	Short code used in reports.
name	VARCHAR(255)	no		Name as the source keeps it.
legal_name	VARCHAR(255)	yes		Legal name.
country_code	CHAR(2)	no		Default IN.
state_code	VARCHAR(8)	yes		GST state code.
gstin	VARCHAR(32)	yes		15 characters when valid; wider so source data that isn't fits (migration 0010).
pan	VARCHAR(32)	yes		Pan.
base_currency_code	CHAR(3)	no	FK	Default INR.
fy_start_month	TINYINT	no		Default 4 (April) (DAT-006).
books_from	DATE	yes		Company inception / books start (IMP-001).
is_active	BOOL	no		Whether it is in use.
extra	JSON	yes		ADR-025
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Refers to: base_currency_code → currencies.code.

Referred to by: account_groups, alerts, audit_log, branches, cost_centres, dimension_values, entity_sources, financial_years, fx_revaluation_snapshots, godowns, inventory_lines, item_groups, items, ledgers, parties, source_fields, statement_lines, stock_values, units, user_entity_access, validation_runs, voucher_lines, voucher_types, vouchers.

ledgerbridge_accounting.entity_sources

Purpose: Which source company (by its key, Tally's GUID) belongs to which entity, on which connection.

Filled by: Transform, with the entity.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.

Column	Type	Null	Key	Meaning
source_entity_key	VARCHAR(191)	no	UK	Tally company GUID or name, Zoho organization ID.
source_entity_name	VARCHAR(255)	no		Source entity name.
precedence	SMALLINT	no		Higher wins on overlap (IMP-005).
active_from	DATE	yes		Active from.
active_to	DATE	yes		Active to.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_entity_key).

Refers to: entity_id → entities.id; connection_id → connections.id.

ledgerbridge_accounting.branches

Purpose: Branches of an entity (planned).

Filled by: Not filled yet.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
code	VARCHAR(64)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
state_code	VARCHAR(8)	yes		GST state code.
gstin	VARCHAR(32)	yes		GST registration number.
is_active	BOOL	no		Whether it is in use.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (entity_id, code).

Refers to: entity_id → entities.id.

Referred to by: user_entity_access, vouchers.

ledgerbridge_accounting.financial_years

Purpose: Financial years of an entity (planned).

Filled by: Not filled yet.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
label	VARCHAR(64)	no		Label.
start_date	DATE	no	UK	Start date.

Column	Type	Null	Key	Meaning
end_date	DATE	no		End date.
is_closed	BOOL	no		Whether closed.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (entity_id, start_date).

Refers to: entity_id → entities.id.

ledgerbridge_accounting.currencies

Purpose: Currencies by ISO code.

Filled by: Seeded (INR); foreign currencies in M4b (FX-001).

Column	Type	Null	Key	Meaning
code	CHAR(3)	no	PK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
decimal_places	TINYINT	no		Decimals shown.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Referred to by: entities, exchange_rates, ledgers, vouchers.

ledgerbridge_accounting.exchange_rates

Purpose: Exchange rates by date (outline; redesigned for FX-003).

Filled by: Not filled yet.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
from_code	CHAR(3)	no	FK, UK	Refers to currencies.
to_code	CHAR(3)	no	FK, UK	Refers to currencies.
rate_date	DATE	no	UK	Rate date.
rate	NUMERIC(20, 8)	no		Rate.
source	VARCHAR(64)	yes		Source.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (from_code, to_code, rate_date).

Refers to: from_code → currencies.code; to_code → currencies.code.

`ledgerbridge_accounting.standard_accounts`

Purpose: The standard chart of accounts used to compare and combine companies (RPT-006).

Filled by: Seeded by a migration.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`parent_id`	BIGINT UNSIGNED	yes	FK	Hierarchy.
`code`	VARCHAR(32)	no	UK	Short stable code.
`name`	VARCHAR(255)	no		Name as the source keeps it.
`nature`	ENUM('asset','liability','equity','income','expense')	no		Nature.
`statement`	ENUM('balance_sheet','profit_loss')	no		Statement.
`statement_line`	VARCHAR(64)	no		e.g. `revenue`, `cogs`, `current_assets`.
`cash_flow_class`	ENUM('operating','investing','financing','cash','none')	no		For cash flow statement.
`sort_order`	INTEGER	no		Sort order.
`created_at`	DATETIME(6)	no		When the row was created (UTC).
`updated_at`	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Refers to: `parent_id` → `standard_accounts.id`.

Referred to by: `account_groups`, `ledgers`, `standard_accounts`.

`ledgerbridge_accounting.account_groups`

Purpose: The source's group tree (Tally groups), per entity, as kept there.

Filled by: Transform, from group masters; renamed reserved groups keep their reserved name as an alias.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`entity_id`	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
`parent_id`	BIGINT UNSIGNED	yes	FK	The parent row in the same table (a tree).
`name`	VARCHAR(255)	no		Name as the source keeps it.
`nature`	ENUM('asset','liability','equity','income','expense')	yes		Derived from primary group.
`is_primary`	BOOL	no		Tally's 28 predefined primary groups.
`affects_gross_profit`	BOOL	no		Tally flag for P&L layout.
`standard_account_id`	BIGINT UNSIGNED	yes	FK	Default mapping for ledgers under this group.
`source_key`	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.

Column	Type	Null	Key	Meaning
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_account_groups_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `parent_id` → `account_groups.id`; `standard_account_id` → `standard_accounts.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `account_groups`, `ledgers`.

`ledgerbridge_accounting.ledgers`

Purpose: The source's ledgers, per entity, with their group, kind, standard account, opening balance as at the books' beginning and tax type.

Filled by: Transform, from ledger masters fetched for a period starting on the books' beginning.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
group_id	BIGINT UNSIGNED	no	FK	Refers to `account_groups`.
name	VARCHAR(255)	no		Name as the source keeps it.
alias	VARCHAR(255)	yes		Alias.
ledger_kind	ENUM('party','bank','cash','tax','stock','income','expense','asset','liability','equity','other')	no		party, bank, cash, tax, stock, … (from the reserved group).
party_id	BIGINT UNSIGNED	yes	FK	Set for customer/vendor ledgers.
standard_account_id	BIGINT UNSIGNED	yes	FK	Overrides group mapping.

Column	Type	Null	Key	Meaning
mapping_status	ENUM('auto','suggested','confirmed','unmapped')	no		auto (a standard account was found) or unmapped.
opening_balance	NUMERIC(20, 4)	no		Signed; as at opening_balance_date.
opening_balance_date	DATE	yes		Opening balance date.
currency_code	CHAR(3)	no	FK	Currency (ISO code).
is_bill_wise	BOOL	no		Whether bill wise.
is_cost_centre_applicable	BOOL	no		Whether cost centre applicable.
gstin	VARCHAR(32)	yes		GST registration number.
tax_type	ENUM('cgst','sgst','igst','cess','tds','tcs','vat','other')	yes		cgst, sgst, igst, cess, tds, tcs, vat for tax ledgers.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_ledgers_entity_group` (entity_id, group_id); index `ix_ledgers_entity_name` (entity_id, name); index `ix_ledgers_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `group_id` → `account_groups.id`; `party_id` → `parties.id`; `standard_account_id` → `standard_accounts.id`; `currency_code` → `currencies.code`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `fx_revaluation_snapshots`, `voucher_lines`.

ledgerbridge_accounting: masters

ledgerbridge_accounting.parties

Purpose: Customers and vendors (the ledgers under Sundry Debtors / Creditors), with GST, PAN and contact details (sensitive: AI-015).

Filled by: Transform, with their ledgers.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
name	VARCHAR(255)	no		**Sensitive** (masking candidate).
party_type	ENUM('customer','vendor','both','other')	no		Tally: Sundry Debtors → customer, Sundry Creditors → vendor.
gstin	VARCHAR(32)	yes		GST registration number.
pan	VARCHAR(32)	yes		**Sensitive.**
state_code	VARCHAR(8)	yes		GST state code.
country_code	CHAR(2)	yes		Country code.
credit_days	INTEGER	yes		Used for ageing and overdue flags.
credit_limit	NUMERIC(20, 4)	yes		Credit limit.
email	VARCHAR(255)	yes		E-mail address.
phone	VARCHAR(255)	yes		Phone.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index ix_parties_raw_record (raw_record_id).

Refers to: entity_id → entities.id; connection_id → connections.id; import_batch_id → import_batches.id.

Referred to by: bill_allocations, ledgers, vouchers.

ledgerbridge_accounting.items

Purpose: Stock items with their opening quantity and value.

Filled by: Transform, from stock item masters (as at the books' beginning).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
item_group_id	BIGINT UNSIGNED	yes	FK	Refers to item_groups.
name	VARCHAR(255)	no		Name as the source keeps it.
code	VARCHAR(255)	yes		Short stable code.
unit_id	BIGINT UNSIGNED	yes	FK	Refers to units.
hsn_code	VARCHAR(32)	yes		Hsn code.
gst_rate	NUMERIC(20, 6)	yes		Gst rate.
opening_qty	NUMERIC(20, 6)	no		Opening qty.
opening_value	NUMERIC(20, 4)	no		Opening value.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index ix_items_raw_record (raw_record_id).

Refers to: entity_id → entities.id; item_group_id → item_groups.id; unit_id → units.id; connection_id → connections.id; import_batch_id → import_batches.id.

Referred to by: inventory_lines.

ledgerbridge_accounting.item_groups

Purpose: Stock groups.

Filled by: Transform, from stock group masters.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
parent_id	BIGINT UNSIGNED	yes	FK	The parent row in the same table (a tree).
name	VARCHAR(255)	no		Name as the source keeps it.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_item_groups_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `parent_id` → `item_groups.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `item_groups`, `items`.

ledgerbridge_accounting.units

Purpose: Units of measure.

Filled by: Transform, from unit masters.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.

Column	Type	Null	Key	Meaning
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
symbol	VARCHAR(255)	no		Symbol.
name	VARCHAR(255)	no		Name as the source keeps it.
decimal_places	TINYINT	no		Decimals shown.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_units_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `inventory_lines`, `items`.

`ledgerbridge_accounting.godowns`

Purpose: Godowns (locations) and whose stock they hold (`holding`, ADR-031).

Filled by: Transform, from godown masters.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
parent_id	BIGINT UNSIGNED	yes	FK	The parent row in the same table (a tree).
name	VARCHAR(255)	no		Name as the source keeps it.

Column	Type	Null	Key	Meaning
`holding`	ENUM('own','ours_with_third_party','third_party_with_us')	no		own / ours_with_third_party (a job worker's) / third_party_with_us.
`source_key`	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
`source_alter_id`	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
`origin`	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
`raw_record_id`	BIGINT UNSIGNED	yes		The raw record it was transformed from.
`is_deleted`	BOOL	no		Deleted in the source (soft delete; kept for history).
`extra`	JSON	yes		Source fields with no column, as JSON (ADR-025).
`connection_id`	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
`import_batch_id`	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
`created_at`	DATETIME(6)	no		When the row was created (UTC).
`updated_at`	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_godowns_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `parent_id` → `godowns.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `godowns`, `inventory_lines`.

`ledgerbridge_accounting.cost_centres`

Purpose: Cost centres and their categories.

Filled by: Transform, from cost centre masters.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`entity_id`	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
`parent_id`	BIGINT UNSIGNED	yes	FK	The parent row in the same table (a tree).
`category`	VARCHAR(255)	yes		Category.
`name`	VARCHAR(255)	no		Name as the source keeps it.

Column	Type	Null	Key	Meaning
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_cost_centres_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `parent_id` → `cost_centres.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `cost_centres`, `line_cost_allocations`.

ledgerbridge_accounting.voucher_types

Purpose: The source's voucher types and the standard type each derives from; whether its vouchers post to the books and move stock.

Filled by: Transform, from voucher type masters (type tree resolved, ADR-027).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
name	VARCHAR(255)	no		Name as the source keeps it.
base_type	ENUM('sales','purchase','payment','receipt','journal','contra','debit_note','credit_note','stock_journal','d			Base type.
is_accounting	BOOL	no		Whether accounting.
is_inventory	BOOL	no		Whether inventory.
standard_type	VARCHAR(64)	yes		Standard type.
moves_stock	BOOL	no		Moves stock.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.

Column	Type	Null	Key	Meaning
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_voucher_types_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `vouchers`.

`ledgerbridge_accounting.dimensions`

Purpose: User-defined reporting dimensions (DAT-005), e.g. a promoted UDF.

Filled by: `promote-field`, from the field catalogue.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(64)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
applies_to	ENUM('voucher','line')	no		Applies to.
is_active	BOOL	no		Whether it is in use.
source_object_type	VARCHAR(64)	yes		Source object type.
source_field	VARCHAR(255)	yes		Source field.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Referred to by: `dimension_values`.

`ledgerbridge_accounting.dimension_values`

Purpose: The values of a dimension.

Filled by: Transform, for promoted fields.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
dimension_id	BIGINT UNSIGNED	no	FK, UK	Refers to dimensions.
entity_id	BIGINT UNSIGNED	yes	FK	The entity (company) the row belongs to.
parent_id	BIGINT UNSIGNED	yes	FK	The parent row in the same table (a tree).
code	VARCHAR(64)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (dimension_id, code).

Refers to: dimension_id → dimensions.id; entity_id → entities.id; parent_id → dimension_values.id.

Referred to by: dimension_values, line_dimensions.

ledgerbridge_accounting: transactions

ledgerbridge_accounting.vouchers

Purpose: Vouchers (transactions) with their type, number, date, party and flags.

Filled by: Transform, from voucher records; replaced as a whole with their child rows (schema.md 8 rule 7).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
branch_id	BIGINT UNSIGNED	yes	FK	Refers to branches.
voucher_type_id	BIGINT UNSIGNED	no	FK	Refers to voucher_types.
voucher_number	VARCHAR(255)	no		The source's voucher number (up to 255 characters).
voucher_date	DATE	no		The voucher's date (denormalised for fast period queries).
reference_number	VARCHAR(255)	yes		The source's reference (e.g. the supplier's invoice number).
reference_date	DATE	yes		Reference date.
effective_date	DATE	yes		The source's effective date (Tally EFFECTIVEDATE); balances count by voucher_date, as Tally does. Migration 0016.

Column	Type	Null	Key	Meaning
party_id	BIGINT UNSIGNED	yes	FK	Main party, if any.
narration	TEXT	yes		Searchable via RAG later (AI-010).
currency_code	CHAR(3)	no	FK	Currency (ISO code).
exchange_rate	NUMERIC(20, 8)	no		Exchange rate.
is_cancelled	BOOL	no		Whether cancelled.
is_optional	BOOL	no		Tally optional/memo vouchers, excluded from books.
is_post_dated	BOOL	no		The source marks it post-dated; it counts from its date.
source_key	VARCHAR(191)	no	UK	The source's own key (Tally: GUID); unique per connection.
source_alter_id	BIGINT	yes		The source's change counter (Tally: AlterID) when last stored.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
raw_record_id	BIGINT UNSIGNED	yes		The raw record it was transformed from.
is_deleted	BOOL	no		Deleted in the source (soft delete; kept for history).
extra	JSON	yes		Source fields with no column, as JSON (ADR-025).
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_key); index `ix_vouchers_entity_date` (entity_id, voucher_date); index `ix_vouchers_entity_type_date` (entity_id, voucher_type_id, voucher_date); index `ix_vouchers_party_date` (party_id, voucher_date); index `ix_vouchers_raw_record` (raw_record_id).

Refers to: `entity_id` → `entities.id`; `branch_id` → `branches.id`; `voucher_type_id` → `voucher_types.id`; `party_id` → `parties.id`; `currency_code` → `currencies.code`; `connection_id` → `connections.id`; `import_batch_id` → `import_batches.id`.

Referred to by: `inventory_lines`, `tax_lines`, `voucher_lines`.

`ledgerbridge_accounting.voucher_lines`

Purpose: The double-entry lines of each voucher (debit +, credit −); `is_effective` says whether a line counts in the books.

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
voucher_id	BIGINT UNSIGNED	no	FK	Refers to vouchers.
entity_id	BIGINT UNSIGNED	no	FK	Denormalized for fast filtering.
voucher_date	DATE	no		Denormalized for fast period queries.
line_no	INTEGER	no		Position of the line in its voucher.
ledger_id	BIGINT UNSIGNED	no	FK	Refers to ledgers.
amount	NUMERIC(20, 4)	no		Base currency, debit +, credit −.
amount_fc	NUMERIC(20, 4)	yes		Foreign-currency amount (Tally: the part before @ of -$100.00 @ Rs82.00/$ = -Rs8200.00); its rate is amount / amount_fc. NULL in the base currency.
is_party_line	BOOL	no		Whether party line.
is_effective	BOOL	no		Counts in the books: not cancelled, optional or deleted, and of a type that posts.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index ix_voucher_lines_entity_ledger_date (entity_id, ledger_id, voucher_date); index ix_voucher_lines_voucher (voucher_id).

Refers to: voucher_id → vouchers.id, on delete cascade; entity_id → entities.id; ledger_id → ledgers.id.

Referred to by: bill_allocations, line_cost_allocations, line_dimensions, tax_lines.

ledgerbridge_accounting.bill_allocations

Purpose: Bill-wise allocations of a line (receivables and payables ageing).

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
voucher_line_id	BIGINT UNSIGNED	no	FK	Refers to voucher_lines.
party_id	BIGINT UNSIGNED	yes	FK	Refers to parties.
bill_type	ENUM('new_ref','against_ref','advance','on_account')	no		Bill type.
bill_name	VARCHAR(255)	yes		Bill name.
bill_date	DATE	yes		Bill date.
due_date	DATE	yes		Due date.
amount	NUMERIC(20, 4)	no		Amount in the base currency, canonical sign (debit +, credit −).

Column	Type	Null	Key	Meaning
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: voucher_line_id → voucher_lines.id, on delete cascade; party_id → parties.id.

ledgerbridge_accounting.line_cost_allocations

Purpose: Cost centre allocations of a line.

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
voucher_line_id	BIGINT UNSIGNED	no	FK	Refers to voucher_lines.
cost_centre_id	BIGINT UNSIGNED	no	FK	Refers to cost_centres.
amount	NUMERIC(20, 4)	no		Amount in the base currency, canonical sign (debit +, credit −).
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: voucher_line_id → voucher_lines.id, on delete cascade; cost_centre_id → cost_centres.id.

ledgerbridge_accounting.line_dimensions

Purpose: A voucher line's value for each dimension.

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
voucher_line_id	BIGINT UNSIGNED	no	PK, FK	Refers to voucher_lines.
dimension_value_id	BIGINT UNSIGNED	no	PK, FK	Refers to dimension_values.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: voucher_line_id → voucher_lines.id, on delete cascade; dimension_value_id → dimension_values.id.

ledgerbridge_accounting.inventory_lines

Purpose: Stock movements of a voucher (in +, out −) by item, godown and batch; stock_quantity is what moves stock (tracked lines move nothing).

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
voucher_id	BIGINT UNSIGNED	no	FK	Refers to vouchers.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
voucher_date	DATE	no		The voucher's date (denormalised for fast period queries).
line_no	INTEGER	no		Position of the line in its voucher.
item_id	BIGINT UNSIGNED	no	FK	Refers to items.
godown_id	BIGINT UNSIGNED	yes	FK	Refers to godowns.
quantity	NUMERIC(20, 6)	no		Quantity.
stock_quantity	NUMERIC(20, 6)	no		What the line moves in stock: the quantity less the part tracked to a note.
unit_id	BIGINT UNSIGNED	yes	FK	Refers to units.
rate	NUMERIC(20, 6)	yes		Rate.
amount	NUMERIC(20, 4)	no		Amount in the base currency, canonical sign (debit +, credit −).
discount_pct	NUMERIC(20, 6)	yes		Discount pct.
batch_name	VARCHAR(255)	yes		Batch name.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: voucher_id → vouchers.id, on delete cascade; entity_id → entities.id; item_id → items.id; godown_id → godowns.id; unit_id → units.id.

ledgerbridge_accounting.tax_lines

Purpose: Tax amounts of a voucher (GST, TDS, …) from lines posted to tax ledgers.

Filled by: Transform, with the voucher.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
voucher_id	BIGINT UNSIGNED	no	FK	Refers to vouchers.
voucher_line_id	BIGINT UNSIGNED	yes	FK	Refers to voucher_lines.
tax_type	ENUM('cgst','sgst','igst','cess','tds','tcs','vat','other')	no		Tax type.
rate	NUMERIC(20, 6)	yes		Rate.
taxable_amount	NUMERIC(20, 4)	yes		Taxable amount.
tax_amount	NUMERIC(20, 4)	no		Tax amount.
hsn_code	VARCHAR(32)	yes		Hsn code.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: voucher_id → vouchers.id, on delete cascade; voucher_line_id → voucher_lines.id, on delete cascade.

ledgerbridge_accounting: field catalogue

`ledgerbridge_accounting.source_fields`

Purpose: Field catalogue (ADR-025): every source field that has no column, kept in `extra`, with counts and an inferred type.

Filled by: Transform.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`entity_id`	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
`object_type`	VARCHAR(64)	no	UK	Kind of source object.
`field_path`	VARCHAR(255)	no	UK	Field path.
`prefix`	VARCHAR(64)	yes		Prefix.
`inferred_type`	ENUM('text','number','date','flag','mixed')			Inferred type.
`record_count`	BIGINT	no		Record count.
`first_seen_at`	DATETIME(6)	no		When: first seen.
`last_seen_at`	DATETIME(6)	no		When: last seen.

Keys and indexes: unique (entity_id, object_type, field_path).

Refers to: `entity_id` → `entities.id`.

ledgerbridge_accounting: source figures

`ledgerbridge_accounting.stock_values`

Purpose: The source's own stock value as of a date (DAT-013, ADR-040): the total its Profit and Loss uses, and by stock group and godown.

Filled by: Validation (as-of date, year ends) and `./dev.ps1 statements` (every month end and year end).

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`entity_id`	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
`connection_id`	BIGINT UNSIGNED	no	FK	The connection the row came from.
`origin`	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
`as_of`	DATE	no	UK	As of.
`scope`	ENUM('total','stock_group','godown')	no	UK	total (what the source's profit uses), stock_group or godown.
`scope_name`	VARCHAR(255)	no	UK	Scope name.
`opening_value`	NUMERIC(20, 4)	yes		Opening value.
`closing_value`	NUMERIC(20, 4)	yes		Closing value.
`integrated`	BOOL	yes		Whether the source integrates accounts with inventory (total rows).
`read_at`	DATETIME(6)	no		When: read.
`created_at`	DATETIME(6)	no		When the row was created (UTC).

Column	Type	Null	Key	Meaning
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (entity_id, as_of, scope, scope_name).

Refers to: entity_id → entities.id; connection_id → connections.id.

ledgerbridge_accounting.fx_revaluation_snapshots

Purpose: Each foreign-currency ledger's balance as the source reports it (revalued at current rates) against its book value (FX-004, ADR-037, ADR-040).

Filled by: Validation and ./dev.ps1 statements, for each date.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.
ledger_id	BIGINT UNSIGNED	no	FK, UK	Refers to ledgers.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.
origin	ENUM('live','backup','file','manual')	no		live or backup (IMP-004).
as_of	DATE	no	UK	Month end, financial year end or validation date (a reliable date in Educational mode, core/dates.py).
currency_code	CHAR(3)	yes		Currency (ISO code).
balance_fc	NUMERIC(20, 4)	yes		The source's own balance in the currency (the foreign part of its closing balance).
book_value	NUMERIC(20, 4)	no		The source's base-currency value of its vouchers (closing balance less the calculated line).
revalued_value	NUMERIC(20, 4)	no		The source's closing balance at current rates.
unrealised_gain_loss	NUMERIC(20, 4)	no		Book value less revalued value (a debit is an unrealised loss).
read_at	DATETIME(6)	no		When: read.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (ledger_id, as_of); index ix_fx_revaluation_entity_date (entity_id, as_of).

Refers to: entity_id → entities.id; ledger_id → ledgers.id; connection_id → connections.id.

ledgerbridge_reporting: statements, ours and the source's (views on top)

ledgerbridge_reporting.statement_lines

Purpose: Trial Balance, Profit & Loss and Balance Sheet lines, ours and the source's, as of each stored date (RPT-012, ADR-040); the views v_trial_balance, v_profit_loss, v_balance_sheet and v_statement_check read them.

Filled by: Validation and `./dev.ps1 statements` (core/statements.py).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
entity_id	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
as_of	DATE	no	UK	As of.
statement	ENUM('trial_balance','profit_loss','balance_sheet')	no	UK	Statement.
figures	ENUM('ours','source')	no	UK	ours (built here) or source (the source's own report).
line_no	INTEGER	no	UK	Position of the line in its voucher.
section	VARCHAR(32)	yes		Section.
label	VARCHAR(255)	no		Label.
match_key	VARCHAR(255)	no		The label as both sides are compared.
level	ENUM('main','sub','detail')	no		main, sub (inside a block) or detail (ours only, never compared).
amount	NUMERIC(20, 4)	no		Amount in the base currency, canonical sign (debit +, credit −).
debit	NUMERIC(20, 4)	yes		Debit.
credit	NUMERIC(20, 4)	yes		Credit.
refreshed_at	DATETIME(6)	no		When: refreshed.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (entity_id, as_of, statement, figures, line_no); index `ix_statement_lines_match` (entity_id, as_of, statement, match_key).

Refers to: `entity_id` → `entities.id`.

ledgerbridge_system: connections and sync

ledgerbridge_system.sources

Purpose: Registered connector types (e.g. `tally`).

Filled by: Seeded on first use.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(32)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
connector_version	VARCHAR(32)	no		Connector version.

Column	Type	Null	Key	Meaning
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Referred to by: connections.

ledgerbridge_system.agents

Purpose: Connector agents near on-premise sources (Phase 1b).

Filled by: Not filled yet.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
name	VARCHAR(255)	no	UK	Name as the source keeps it.
cert_fingerprint	VARCHAR(128)	yes		Cert fingerprint.
version	VARCHAR(32)	yes		Version.
last_seen_at	DATETIME(6)	yes		When: last seen.
status	ENUM('active','disabled')	no		Current state.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (name).

Referred to by: connections.

ledgerbridge_system.connections

Purpose: Each configured connection: source, origin, companies, settings, status.

Filled by: connect; status by connect -Activate / -Paused.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
source_id	BIGINT UNSIGNED	no	FK	Refers to sources.
name	VARCHAR(255)	no	UK	Name as the source keeps it.
config	JSON	no		Config.
credentials_enc	VARBINARY(4096)	yes		Credentials enc.
agent_id	BIGINT UNSIGNED	yes	FK	Refers to agents.
sync_interval_sec	INTEGER	no		Sync interval sec.
status	ENUM('active','paused','error')	no		Current state.
created_by	VARCHAR(36)	yes		Created by.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (name).

Refers to: source_id → sources.id; agent_id → agents.id.

Referred to by: account_groups, alerts, cost_centres, entity_sources, file_layouts, fx_revaluation_snapshots, godowns, import_batches, item_groups, items, ledgers, parties, raw_records, stock_values, sync_checkpoints, sync_logs, sync_markers, sync_runs, sync_slow_periods, sync_unverified_ranges, units, voucher_types, vouchers.

ledgerbridge_system.sync_markers

Purpose: The last AlterID stored per connection, company and object type (incremental syncs).

Filled by: Sync.

Column	Type	Null	Key	Meaning
connection_id	BIGINT UNSIGNED	no	PK, FK	The connection the row came from.
source_entity_key	VARCHAR(191)	no	PK	The source company's key (Tally: company GUID).
object_type	VARCHAR(64)	no	PK	Kind of source object.
marker_value	VARCHAR(191)	no		Marker value.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: connection_id → connections.id.

ledgerbridge_system.sync_runs

Purpose: Every sync, deletion check and transform run with its status and counts (SYN-005).

Filled by: Sync, deletion check, transform.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.
run_type	ENUM('incremental','full','import','deletion_check','transform')	no		Run type.
status	ENUM('running','succeeded','failed','interrupted')	no		Current state.
started_at	DATETIME(6)	no		When it started.
finished_at	DATETIME(6)	yes		When it finished.
records_received	INTEGER	no		Records received.
records_failed	INTEGER	no		Records failed.
error_summary	TEXT	yes		Error summary.
heartbeat_at	DATETIME(6)	yes		When: heartbeat.
pid	INTEGER	yes		Pid.
host	VARCHAR(64)	yes		Host.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: connection_id → connections.id.

Referred to by: alerts, raw_records, sync_checkpoints, sync_unverified_ranges, validation_runs.

`ledgerbridge_system.sync_logs`

Purpose: Output lines of a background job, for `sync watch` (ADR-024).

Filled by: Background jobs.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`connection_id`	BIGINT UNSIGNED	no	FK	The connection the row came from.
`job_id`	CHAR(36)	no		Job id.
`logged_at`	DATETIME(6)	no		When: logged.
`line`	TEXT	no		Line.

Keys and indexes: index `ix_sync_logs_job` (job_id, id).

Refers to: `connection_id` → `connections.id`.

`ledgerbridge_system.sync_checkpoints`

Purpose: Chunks of a full sync fetched and stored, so an interrupted one resumes (ADR-022).

Filled by: Sync.

Column	Type	Null	Key	Meaning
`id`	BIGINT UNSIGNED	no	PK	Row id.
`connection_id`	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
`source_entity_key`	VARCHAR(191)	no	UK	The source company's key (Tally: company GUID).
`object_type`	VARCHAR(64)	no		Kind of source object.
`chunk_key`	VARCHAR(128)	no	UK	e.g. voucher:2023-04-01..2023-04-30
`records`	INTEGER	no		Records.
`sync_run_id`	BIGINT UNSIGNED	yes	FK	The sync run it belongs to.
`created_at`	DATETIME(6)	no		When the row was created (UTC).
`updated_at`	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, source_entity_key, chunk_key).

Refers to: `connection_id` → `connections.id`; `sync_run_id` → `sync_runs.id`.

`ledgerbridge_system.sync_slow_periods`

Purpose: Months the source couldn't deliver in one request; later runs split them directly.

Filled by: Sync.

Column	Type	Null	Key	Meaning
`connection_id`	BIGINT UNSIGNED	no	PK, FK	The connection the row came from.

Column	Type	Null	Key	Meaning
source_entity_key	VARCHAR(191)	no	PK	The source company's key (Tally: company GUID).
object_type	VARCHAR(64)	no	PK	Kind of source object.
month	DATE	no	PK	first day of the month
piece_days	INTEGER	no		Piece days.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: connection_id → connections.id.

ledgerbridge_system.sync_unverified_ranges

Purpose: Date ranges whose records couldn't be proven complete (ADR-023).

Filled by: Sync.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.
source_entity_key	VARCHAR(191)	no		The source company's key (Tally: company GUID).
object_type	VARCHAR(64)	no		Kind of source object.
period_from	DATE	no		Period from.
period_to	DATE	no		Period to.
reason	VARCHAR(255)	no		counts and kinds only, never names
sync_run_id	BIGINT UNSIGNED	yes	FK	The sync run it belongs to.
resolved_at	DATETIME(6)	yes		When it was resolved (NULL while open).
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index ix_sync_unverified_open (connection_id, source_entity_key, resolved_at).

Refers to: connection_id → connections.id; sync_run_id → sync_runs.id.

ledgerbridge_system.import_batches

Purpose: Backup or file imports (IMP-003, IMP-004).

Filled by: Imports (later).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.

Column	Type	Null	Key	Meaning
kind	ENUM('backup','file')	no		Kind.
file_name	VARCHAR(255)	yes		File name.
file_sha256	CHAR(64)	yes		File sha256.
period_from	DATE	yes		Period from.
period_to	DATE	yes		Period to.
status	ENUM('pending','running','succeeded','failed')	no		Current state.
created_by	VARCHAR(36)	yes		Created by.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: connection_id → connections.id.

Referred to by: account_groups, cost_centres, godowns, item_groups, items, ledgers, parties, raw_records, units, validation_runs, voucher_types, vouchers.

ledgerbridge_system.file_layouts

Purpose: Saved CSV/TXT column mappings (CON-003).

Filled by: Not filled yet.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
connection_id	BIGINT UNSIGNED	no	FK, UK	The connection the row came from.
name	VARCHAR(255)	no	UK	Name as the source keeps it.
column_mapping	JSON	no		Column mapping.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (connection_id, name).

Refers to: connection_id → connections.id.

ledgerbridge_system: validation and alerts

ledgerbridge_system.validation_runs

Purpose: Each validation of an entity as of a date, with counts (VAL-001).

Filled by: Validation after every sync and on validate.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
sync_run_id	BIGINT UNSIGNED	yes	FK	The sync run it belongs to.
import_batch_id	BIGINT UNSIGNED	yes	FK	The import it came from, for imported data.
entity_id	BIGINT UNSIGNED	no	FK	The entity (company) the row belongs to.

Column	Type	Null	Key	Meaning
as_of	DATE	no		As of.
started_at	DATETIME(6)	no		When it started.
status	ENUM('running','passed','failed')	no		Current state.
checks_passed	INTEGER	no		Checks passed.
checks_failed	INTEGER	no		Checks failed.
thorough	BOOL	yes		Whether the source's controls (counts, stock, bills, year ends) were compared.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: sync_run_id → sync_runs.id; import_batch_id → import_batches.id; entity_id → entities.id.

Referred to by: alerts, validation_issues.

ledgerbridge_system.validation_issues

Purpose: Each failed check of a run, with its code and the two figures; resolved when a later run no longer finds it.

Filled by: Validation.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
validation_run_id	BIGINT UNSIGNED	no	FK	Refers to validation_runs.
check_code	VARCHAR(64)	no		The check (see the error catalogue, LB-VAL-xxx).
severity	ENUM('error','warning','info')	no		Severity.
object_type	VARCHAR(64)	yes		Kind of source object.
object_id	VARCHAR(64)	yes		Object id.
expected_value	NUMERIC(20, 4)	yes		Expected value.
actual_value	NUMERIC(20, 4)	yes		Actual value.
message	TEXT	no		Message.
resolved_at	DATETIME(6)	yes		When it was resolved (NULL while open).
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: validation_run_id → validation_runs.id.

ledgerbridge_system.alerts

Purpose: In-app alerts: one open per connection and kind (SYN-006, ADR-019).

Filled by: Sync and validation.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
kind	ENUM('sync_failed','deletion_check_failed','validation_failed')	no		Kind.

Column	Type	Null	Key	Meaning
severity	ENUM('error','warning')	no		Severity.
connection_id	BIGINT UNSIGNED	no	FK	The connection the row came from.
entity_id	BIGINT UNSIGNED	yes	FK	The entity (company) the row belongs to.
sync_run_id	BIGINT UNSIGNED	yes	FK	The sync run it belongs to.
validation_run_id	BIGINT UNSIGNED	yes	FK	Refers to validation_runs.
message	TEXT	no		Message.
resolved_at	DATETIME(6)	yes		When it was resolved (NULL while open).
acknowledged_at	DATETIME(6)	yes		When: acknowledged.
acknowledged_by	VARCHAR(36)	yes		Acknowledged by.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index ix_alerts_open (connection_id, kind, resolved_at).

Refers to: connection_id → connections.id; entity_id → entities.id; sync_run_id → sync_runs.id; validation_run_id → validation_runs.id.

ledgerbridge_system: users, roles and access

ledgerbridge_system.users

Purpose: Users (same id as the control plane); local password only with AUTH_MODE=local.

Filled by: M5.

Column	Type	Null	Key	Meaning
id	CHAR(36)	no	PK	same ID as the control plane
email	VARCHAR(255)	no	UK	E-mail address.
display_name	VARCHAR(255)	no		Display name.
status	ENUM('active','disabled','locked')	no		Current state.
password_hash	VARCHAR(255)	yes		Password hash.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (email).

Referred to by: user_entity_access, user_roles.

ledgerbridge_system.roles

Purpose: Roles.

Filled by: M5 (seeded defaults).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(64)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.
is_system	BOOL	no		Whether system.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Referred to by: masking_rules, role_permissions, user_roles.

ledgerbridge_system.permissions

Purpose: Permission codes, e.g. validation.view.

Filled by: M5 (seeded).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(64)	no	UK	Short stable code.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

Referred to by: role_permissions.

ledgerbridge_system.role_permissions

Purpose: Which role has which permission.

Filled by: M5.

Column	Type	Null	Key	Meaning
role_id	BIGINT UNSIGNED	no	PK, FK	Refers to roles.
permission_id	BIGINT UNSIGNED	no	PK, FK	Refers to permissions.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: role_id → roles.id; permission_id → permissions.id.

ledgerbridge_system.user_roles

Purpose: Which user has which role.

Filled by: M5.

Column	Type	Null	Key	Meaning
user_id	CHAR(36)	no	PK, FK	Refers to users.
role_id	BIGINT UNSIGNED	no	PK, FK	Refers to roles.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: user_id → users.id; role_id → roles.id.

ledgerbridge_system.user_entity_access

Purpose: Which entities (and branches) a user may see (ACC-002).

Filled by: M5.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
user_id	CHAR(36)	no	FK, UK	Refers to users.
entity_id	BIGINT UNSIGNED	no	FK, UK	The entity (company) the row belongs to.
branch_id	BIGINT UNSIGNED	yes	FK, UK	Refers to branches.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (user_id, entity_id, branch_id).

Refers to: user_id → users.id; entity_id → entities.id; branch_id → branches.id.

ledgerbridge_system.masking_rules

Purpose: Masking per role for sensitive fields (ACC-003, AI-015).

Filled by: M5 / M8.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
role_id	BIGINT UNSIGNED	no	FK	Refers to roles.
target	VARCHAR(64)	no		e.g. party.name, party.pan
mask_type	ENUM('hide','partial','hash')	no		Mask type.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Refers to: role_id → roles.id.

ledgerbridge_system: audit and AI log

ledgerbridge_system.audit_log

Purpose: Append-only audit trail (AUD-001).

Filled by: API and workers (M5).

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
occurred_at	DATETIME(6)	no		When: occurred.
user_id	CHAR(36)	yes		User id.
action	VARCHAR(64)	no		Action.
object_type	VARCHAR(64)	yes		Kind of source object.
object_id	VARCHAR(64)	yes		Object id.
entity_id	BIGINT UNSIGNED	yes	FK	The entity (company) the row belongs to.
details	JSON	yes		Details.
ip_address	VARCHAR(45)	yes		Ip address.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index `ix_audit_log_occurred_at` (occurred_at); index `ix_audit_log_user_id` (user_id).

Refers to: `entity_id` → `entities.id`.

`ledgerbridge_system.ai_query_log`

Purpose: One row per AI question (AI-007).

Filled by: M8.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
user_id	CHAR(36)	yes		User id.
question	TEXT	no		Question.
generated_sql	TEXT	yes		Generated sql.
status	ENUM('succeeded','failed','refused')	no		Current state.
row_count	INTEGER	yes		Row count.
duration_ms	INTEGER	yes		Duration ms.
provider	VARCHAR(64)	yes		Provider.
model	VARCHAR(128)	yes		Model.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: index `ix_ai_query_log_user_created` (user_id, created_at).

`ledgerbridge_system`: configuration

`ledgerbridge_system.kpi_definitions`

Purpose: Business health indicators and thresholds (RPT-002).

Filled by: M6.

Column	Type	Null	Key	Meaning
id	BIGINT UNSIGNED	no	PK	Row id.
code	VARCHAR(64)	no	UK	Short stable code.
name	VARCHAR(255)	no		Name as the source keeps it.

Column	Type	Null	Key	Meaning
formula_view	VARCHAR(128)	no		Formula view.
good_threshold	NUMERIC(20, 4)	yes		Good threshold.
bad_threshold	NUMERIC(20, 4)	yes		Bad threshold.
direction	ENUM('higher_better','lower_better')	no		Direction.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

Keys and indexes: unique (code).

`ledgerbridge_system.settings`

Purpose: Key-value settings.

Filled by: Configuration.

Column	Type	Null	Key	Meaning
key	VARCHAR(128)	no	PK	Key.
value	JSON	no		Value.
created_at	DATETIME(6)	no		When the row was created (UTC).
updated_at	DATETIME(6)	no		When the row was last changed (UTC).

ledgerbridge_reporting

No tables yet: the reporting views and aggregate tables are built in M6 (designed in schema.md Section 6).